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PROJECT REPORT

CSE0001 – Digital Literacy

Student Digital Ambassador Portfolio

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Branch: Artificial Intelligence

Year: First Year (2025–2026)

Course Code: CSE0001

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INTRODUCTION

This report documents my work as a Student Digital Ambassador for the CSE0001 Digital Literacy course at VIT Bhopal University. The project is structured around a realistic scenario in which I, as a newly appointed Digital Ambassador, must build a comprehensive Digital Literacy Portfolio to guide my fellow first-year students through the digital landscape. Over the course of five tasks, I have created an awareness infographic, set up professional online profiles, explored coding and collaboration platforms, drafted professional emails with an etiquette guide, and researched cybercrime threats with a focus on the Colonial Pipeline ransomware attack. Each task has been completed using free, browser-based tools, and the entire body of work is organised in a public GitHub repository. This report presents my reflections, observations, and key learnings from each task.

TASK 1 – DIGITAL LITERACY AWARENESS INFOGRAPHIC

For Task 1, I used Canva — a free, browser-based graphic design platform — to create a one-page digital literacy awareness infographic for my fellow first-year students at VIT Bhopal University. Canva's drag-and-drop interface and its extensive library of templates made it possible to produce a visually appealing design without any prior graphic design experience, aligning well with the spirit of this assignment.

My infographic covers three core topics. First, it explains what digital literacy is and why it matters in today's academic and professional world. Second, it highlights useful digital tools for students, including Google Workspace for collaboration, GitHub for code management, and HackerRank for skill development. Third, it addresses safe internet practices, covering strong password creation, recognising phishing emails, and responsible social media conduct. The content is organised into clearly labelled sections with icons, bold headings, and a consistent colour palette.

The most interesting challenge was balancing the volume of information on a single page. Initially I tried to include too many topics, resulting in a cluttered layout. I had to make deliberate editorial decisions — selecting only the most impactful points and expressing complex ideas in short statements paired with visuals. This taught me that effective communication is as much about what you leave out as what you include.

TASK 2 – STUDENT DIGITAL PORTFOLIO

For Task 2, I set up professional accounts on three key platforms: GitHub, LinkedIn, and Kaggle. Each serves a distinct purpose, and together they form the foundation of a credible long-term digital identity that I will build upon throughout my four years at VIT Bhopal University.

GitHub is a version control and code-hosting platform used by developers and researchers worldwide. I created a profile README repository containing my name, branch, year, and a statement about what I hope to learn during my programme. GitHub will serve as my living portfolio of code, academic projects, and collaborative contributions throughout my degree.

LinkedIn is the world's largest professional networking platform. I completed my Education section with my current degree, institution (VIT Bhopal University), and expected graduation year. Over the next four years I plan to add skills, certifications, and project experiences, and use the platform to connect with industry professionals and apply for internships and graduate roles.

Kaggle is a data science and machine learning community where learners participate in competitions, share datasets, and publish notebooks. As a student of Artificial Intelligence at VIT Bhopal, Kaggle is directly aligned with my field. I plan to participate in beginner competitions and study public notebooks to build my modelling and analysis skills, aiming for a strong profile by my third year.

TASK 3 – CODING & COLLABORATION PLATFORMS

For Task 3, I worked with HackerRank for the coding practice component and Google Forms for the collaborative tool-building component.

I created an account on ***HackerRank***, a competitive programming and skill assessment platform trusted by leading technology companies for technical screening. I completed the beginner-level "Solve Me First" challenge, which involves reading two integers and printing their sum. While the logic is elementary, navigating the online judge — reading the problem statement carefully, understanding input/output format requirements, and submitting a clean solution — was a valuable introduction to competitive programming. HackerRank also provides structured skill tracks in Python, data structures, SQL, and algorithms, which I plan to work through during my first and second years.

For the Google Workspace component, I created a five-question Digital Literacy Awareness Quiz using Google Forms, including multiple choice and short answer questions. The form is linked to a Google Sheet that automatically captures all responses, making it easy to analyse results. This gave me hands-on experience with tools used in academic research and collaborative projects, and the quiz itself can serve as a real awareness resource for my batchmates.

TASK 4 – PROFESSIONAL EMAIL & ETIQUETTE GUIDE

Part A – Professional Email Drafts

Email 1: Assignment Extension Request

Subject: Request for Assignment Submission Extension – Sanu Singh – 25MIM10171

Dear Professor,

I hope you are doing well.

I am writing to respectfully request an extension for the upcoming assignment due on 31st March 2026. Unfortunately, I have been dealing with High Fever, which has made it difficult for me to complete the work to the best of my ability within the given timeframe.

I sincerely value this course and want to ensure that I submit an assignment that reflects my understanding and effort. Therefore, I kindly request an extension until 5th April 2026.

I apologize for any inconvenience caused and appreciate your understanding and consideration. Please let me know if any documentation or further information is required.

Thank you for your time.

Yours sincerely,

Sanu Singh

Int. M.Tech, Artificial Intelligence

25MIM10171

Email 2: Summer Internship Interest

Subject: Application for Summer Internship Opportunity – Sanu Singh

Dear Internship Coordinator,

I hope this message finds you well.

I am writing to express my interest in applying for a summer internship opportunity at your esteemed organization. I am currently a 1st year student pursuing Integrated M.Tech in Artificial Intelligence at VIT Bhopal University.

I am particularly interested in gaining practical experience in Software Development, and I believe that an internship with your organization would provide valuable exposure and learning opportunities. I am eager to contribute my skills, including Python Automation, Data Cleaning, and Bug Fixing, while also learning from experienced professionals.

I have attached my resume for your review. I would be grateful for the opportunity to discuss how I can contribute to your team.

Thank you for considering my application. I look forward to your response.

Yours sincerely,

Sanu Singh

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Part B – Responsible Social Media Checklist

Do's

- **Update Privacy Settings:** Regularly review who can see your posts and personal info.
- **Network Professionally:** Use platforms like LinkedIn to highlight your skills in Python and Data Science.
- **Think Before Posting:** Assume everything you post is permanent and could be seen by a future employer.
- **Fact-Check Content:** Verify information from reliable sources before hitting "Share."
- **Secure Your Accounts:** Enable 2FA to prevent unauthorized access to your identity.

Don'ts

- **Share Sensitive Info:** Never post your address, student ID, or private contact details.
- **Post During Conflict:** Avoid posting while angry; digital footprints are hard to erase.

- **Plagiarize Content:** Never share others' work or AI-generated content without proper attribution.
- **Ignore Terms of Service:** Be aware of how platforms use your data.
- **Post Compromising Content:** Avoid sharing anything that violates college conduct or professional ethics.

Reflection – Poor Digital Communication Scenario

For the reflection component of Task 4, I considered a scenario that illustrates vividly how poor digital communication can escalate a simple academic concern into a serious institutional matter. A student who received a lower-than-expected mid-semester grade sent an angry email to their professor using confrontational language, no formal greeting, no sign-off, and block capitals throughout — widely interpreted as aggressive in digital communication. The professor forwarded the email to the department head and academic advisor, turning a legitimate query into a formal disciplinary discussion.

The situation could have been avoided entirely had the student waited before writing, used a neutral and informative subject line, stated the concern factually, and requested a meeting at the professor's convenience — signed off with their full name and registration number. This reinforced for me that digital communications are permanent records, and that tone and structure are just as important as content in any professional or academic context.

TASK 5 – CYBERCRIME AWARENESS: RANSOMWARE

Part A – Case Study: Colonial Pipeline Ransomware Attack (2021)

For Task 5, I researched the 2021 Colonial Pipeline Ransomware Attack — one of the most consequential cybercrime incidents in recent history. Colonial Pipeline operates approximately 5,500 miles of fuel pipeline, supplying around 45% of the U.S. East Coast's fuel (gasoline, diesel, and jet fuel), and was classified as critical national infrastructure by the U.S. Department of Homeland Security.

On 7 May 2021, the company discovered its IT systems had been compromised by DarkSide ransomware — a Ransomware-as-a-Service (RaaS) group. Attackers gained initial access using a leaked VPN password found on the dark web from a prior breach; crucially, the compromised account had no multi-factor authentication. Threat actors then mapped the internal network using Cobalt Strike, exfiltrated approximately 100 GB of sensitive data in around two hours, and deployed DarkSide ransomware (Salsa20 + RSA-1024 encryption) across billing systems, IT servers, and admin workstations.

Management proactively shut down all 5,500 miles of pipeline operations. Seventeen states declared a state of emergency. CEO Joseph Blount authorised payment of 75 Bitcoin (approximately US\$4.4 million) to obtain the decryption key. The FBI later seized 63.7 Bitcoin (~US\$2.3M) from a DarkSide wallet. The incident led directly to Executive Order 14028 on national cybersecurity and mandatory CISA incident reporting for critical infrastructure operators.

Part B – Stay Safe Online: Prevention Checklist

General Safety Tips

- Use strong and unique passwords for all accounts — avoid reusing the same password everywhere.
- Enable Two-Factor Authentication (2FA) on email, social media, and banking apps.
- Do not click on suspicious links received via email, SMS, or messaging apps.
- Always verify the sender before sharing personal or sensitive information.
- Avoid using public Wi-Fi for logging into important accounts or making transactions.

- Keep your phone, laptop, and apps updated to protect against security vulnerabilities.
- Log out from accounts when using shared or public computers (library, labs, cyber cafes).
- Download apps only from trusted sources like the Google Play Store or Apple App Store.

Financial / UPI Safety Tips

- Never share your UPI PIN, OTP, or banking details with anyone — even if they claim to be from your bank.
- Always check the recipient's name before approving any UPI Collect Request.
- Do not scan unknown QR codes, as they may lead to fraudulent payments.
- Avoid saving card or banking details on shared or public devices.

Reporting Cybercrime

- Report cybercrime incidents immediately on the National Cyber Crime Portal: <https://cybercrime.gov.in>
- Call the Cyber Crime Helpline: 1930 (available 24x7 for financial fraud and urgent help).

Reflection

What surprised me most while researching this topic was the scale of real-world impact from a single cybersecurity lapse. The Colonial Pipeline attack began with one unprotected VPN credential found on the dark web — no sophisticated exploit, no elaborate hacking — just a password without multi-factor authentication. That single oversight shut down fuel supply across 17 states, triggered panic-buying at petrol stations, and cost the company over US\$4.4 million in ransom and an estimated US\$15–20 million in total incident costs. I had assumed ransomware primarily targeted careless individuals; discovering that it can cripple national infrastructure through a basic security failure was a genuine wake-up call.

As a result of this research, the one habit I will personally change is consistently enabling multi-factor authentication on every account I create — from my college email and GitHub to UPI apps and LinkedIn. Before this assignment, I treated 2FA as optional. I now understand it as the single most effective barrier between an attacker and my digital identity.

CONCLUSION

Completing this Digital Literacy Portfolio has been one of the most practically valuable exercises of my first year at VIT Bhopal University. Unlike conventional coursework that ends with a written examination, every task in this project produced something real, usable, and permanently visible in my digital portfolio. I now have an active GitHub repository, professional profiles on LinkedIn and Kaggle, a completed HackerRank challenge, polished professional email drafts, a social media etiquette checklist, a detailed case study on the Colonial Pipeline ransomware attack, and an actionable online safety guide. More significantly, the project has fundamentally shifted my mindset: I now think of my digital presence as something I am actively and deliberately building. The skills and habits developed here — professional identity management, structured digital communication, regular coding practice, enabling 2FA on all accounts, and maintaining offline backups — will remain relevant and compound in value throughout my career in Artificial Intelligence.

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